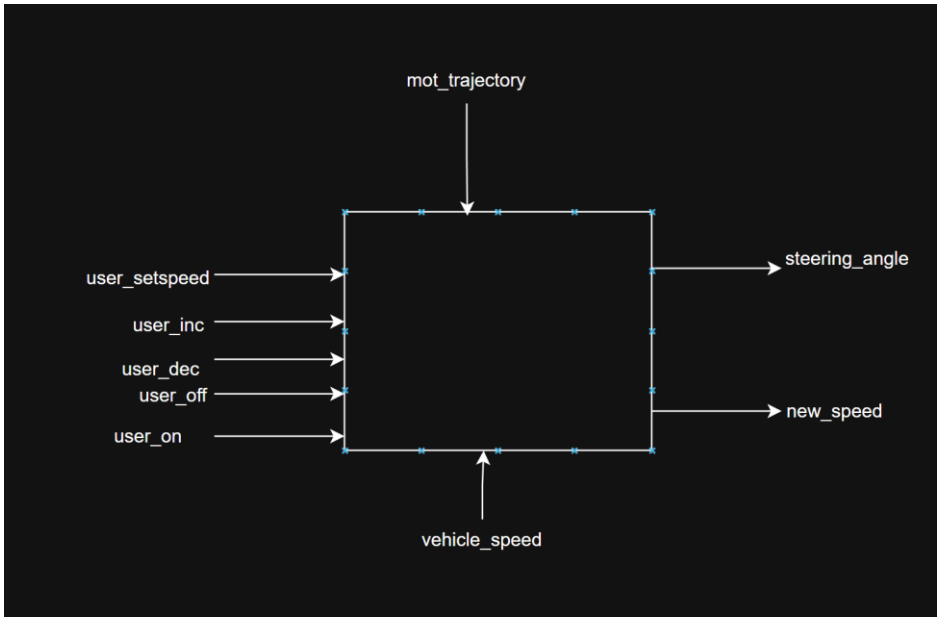
ISE 350 Milestone 1 – Autonomous Vehicle Control

- What information do you need from Motion Planning?
    - The calculated motion trajectory (angle, speed)
- Do you need information about the current vehicle state?
    - Yes, we need its motion and the steering angle
- What information will you send to the motor?
    - The user input (on, off, inc, dec, set_speed)
- What information will you send to the steering mechanism?
    - The motion planning team's calculated trajectory
- What happens when the Safety team sends an emergency-stop command?
    - The subsystem should enter mode off/stop and perform necessary cleanup/handling.
- What operating modes might the controller have?
    - Normal
    - Stop
    - Emergency Stop
    - Error

1. Subsystem Diagram



2. Team Responsibilities:

Student	Initial Responsibility
Holden	Speed control and cmd_speed interface
Reagan	Steering/path tracking and cmd_steering_angle interface
Kael	Operating-mode and safety interface logic
Mason	Simulink integration, testing, and interface documentation

3. I/O Table:

Inbound Signals

Signal 1: desired_speed

- Signal Name: desired_speed
- Source Subsystem: Motion Planning

- Destination Subsystem: Autonomous Vehicle Control
- Description: Desired forward vehicle speed generated by Motion Planning that the speed controller should achieve
- Data Type: single
- Units: m/s
- Expected Range: 0 to 3.5 m/s (TBD – confirm w/ Motion Planning)
- Update Frequency: 50Hz (TBD – coonfirm w/ Motion Planning)
- Coordinate / Reference Frame: Vehicle Frame (+X forward projection)
- Invalid / Fault Value: NaN, < 0.0, or > 3.5 m/s
- Behavior if Signal is Unavailable: Command vehicle speed to 0 m/s and enter a safe Stop/Fault mode

Signal 2: desired_steering

- Signal Name: desired_steering
- Source Subsystem: Motion Planning
- Destination Subsystem: Autonomous Vehicle Control
- Description: Desired forward vehicle speed generated by Motion Planning that the speed controller should achieve
- Data Type: single
- Units: radians
- Expected Range: approx ±25 degrees (-0.436 to +0.436 rad) (TBD – confirm w/ Motion Planning)
- Update Frequency: 50Hz (TBD – coonfirm w/ Motion Planning)
- Coordinate / Reference Frame: Virtual single front wheel (CCW / Left turn = positive)
- Invalid / Fault Value: NaN or value outside the allowable steering range
- Behavior if Signal is Unavailable: Command steering toward 0 rad / straight and enter a safe Stop/Fault mode

Signal 3: fbk_wheel_speed

- Signal Name: fbk_wheel_speed (feedback wheel speed)
- Source Subsystem: Hardware Integration & Active Safety
- Destination Subsystem: Autonomous Vehicle Control, Perception & Sensor Fusion
- Description: Measured vehicle ground speed derived from optical or Hall-effect wheel encoder pulse frequencies.
- Data Type: single

- Units: m/s
- Expected Range: 0.0 to 4.5 m/s
- Update Frequency: 100 Hz, 10ms
- Coordinate / Reference Frame: Longitudinal chassis axis (+X forward)
- Invalid / Fault Value: NaN, < 0.0, or pulse-rate accumulator overflow

- Behavior if Signal is Unavailable: Control and Perception fall back to visual/optical or motor open-loop back-EMF estimates?

Signal 4: fbk_actual_steering

- Signal Name: fbk_actual_steering
- Source Subsystem: Hardware Integration & Active Safety
- Destination Subsystem: Autonomous Vehicle Control
- Description: Modeled actual road wheel steering angle accounting for servo actuation dynamics, calibration offsets, and mechanical rate limits.
- Data Type: single
- Units: rad or degrees
- Expected Range: Approx 25 degrees (-0.436 to 0.436 rad)
- Update Frequency: 50 Hz, 20 ms
- Coordinate / Reference Frame: Front axle center (CCW / Left turn = positive)
- Invalid / Fault Value: NaN,
- Behavior if Signal is Unavailable: Control assumes an idealized zero-lag actuator response model.

Signal 5: safety_supervisory_state

- Signal Name: safety_supervisory_state
- Source Subsystem: Hardware Integration & Active Safety
- Destination Subsystem: Autonomous Vehicle Control, Motion Planning, Perception & Sensor Fusion
- Description: Current operating state of the Stateflow safety supervisor (0 = SYS_INIT_STANDBY, 1 = SYS_NOMINAL_ACTIVE, 2 = SYS_AEB_INTERVENTION, 3 = SYS_ESTOP_LATCHED).
- Data Type: uint8 (Enum)
- Units: N/A (State Enumeration)
- Expected Range: 0 to 3
- Update Frequency: 100 Hz, 10ms

- Coordinate / Reference Frame: N/A
- Invalid / Fault Value: 255 (STATE_FAULT) or any unmapped integer

- Behavior if Signal is Unavailable: Upstream nodes treat safety supervisor as offline and immediately suspend autonomous driving modes.

Signal 6: hw_hardware_health

- Signal Name: hw_hardware_health
- Source Subsystem: Hardware Integration & Active Safety
- Destination Subsystem: Autonomous Vehicle Control, Motion Planning, Perception & Sensor Fusion
- Description: 16-bit status word containing diagnostic health flags for low-level electronics (Bit 0: E-Stop Tripped, Bit 1: Low Battery, Bit 2: ESC Fault, Bit 3: Steering Stall, Bit 4: Watchdog Timeout).
- Data Type: uint16 (Bitmask)
- Units: N/A (Binary bit flags)
- Expected Range: 0x0000 to 0xFFFF
- Update Frequency: 10 Hz, 100 ms
- Coordinate / Reference Frame: N/A
- Invalid / Fault Value: Non-zero critical fault bits asserted
- Behavior if Signal is Unavailable: Upstream software halts trajectory execution and signals operator intervention.

Outbound Signals

Signal 7: cmd_speed

- Signal Name: cmd_speed
- Source Subsystem: Autonomous Vehicle Control
- Destination Subsystem: Hardware Integration & Active Safety
- Description: Demanded forward velocity command generated by the speed controller.
- Data Type: single (or double)
- Units: m/s
- Expected Range: 0.0 to 3.5 m/s
- Update Frequency: 20 ms, 50 Hz
- Coordinate / Reference Frame: Vehicle Frame (ISO 8855, +X forward)

- Invalid / Fault Value: NaN, < 0.0 or > 3.5
- Behavior if Signal is Unavailable: Clamp commanded speed to 0 m/s within 100 ms (smooth down to stop).

Signal 8: cmd_steering_angle

- Signal Name: cmd_steering_angle
- Source Subsystem: Autonomous Vehicle Control
- Destination Subsystem: Hardware Integration & Active Safety
- Description: Demanded road wheel steering angle generated by the path tracking controller.
- Data Type: single (or double)
- Units: rad/s or degrees?
- Expected Range: Approx 25 degrees (-0.436 to 0.436 rad)
- Update Frequency: 50 Hz,
- Coordinate / Reference Frame: Virtual single front wheel (CCW / Left turn = positive)
- Invalid / Fault Value: NaN, <-0.5 or >+0.5
- Behavior if Signal is Unavailable: Neutral lock, set to 0 rad, wheels commanded straight).

4. Team Dependencies

Perception: We need the vehicle's speed and position as inputs to our ACC and lane keeping. We will use these to calculate the instructions for the hardware to maintain speed/lane position.
Motion Planning: We need the motion planning's outputted motion trajectory so we can calculate the steering angle needed to maintain lane position
Hardware: We will send the speed and angle so it can actuate the motors and steering hardware. We will also receive any emergency stops or safety information from this team so that our system can properly respond.

5. Initial Testing Plan

Our group will test different aspects of autonomous vehicle control in Simulink by using simulated data such as desired speed and wheel speed and monitor the output cmd_speed to see if it matches the proper thrust needed. We can do something similar

to test if our system goes into emergency stop mode by sending made up input from the safety team. We can then follow up and see if the system takes the correct steps to make itself stable again. Following tests can be more extreme in sending made up data to see how the system reacts in all situations for bug fixing.